Decomposing Systematic Paired-Rating Disagreement in ELL Writing Assessment across Holistic and Analytic Scores

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Introduction

Human raters do not always assign the same score to the same essay, even when using a common rubric (Lim, 2019). Prior research has linked rater disagreement to essay characteristics, scoring difficulty, and annotator information, but less is known about whether these patterns extend across scoring dimensions, prompts, and rater-pair configurations. Rather than combining two ratings into a single score, the current study treats disagreement itself as a substantive outcome and examines whether paired human-rater disagreement can be characterized, predicted, and generalized across the following seven scoring dimensions: 1) Overall, 2) Cohesion, 3) Syntax, 4) Vocabulary, 5) Phraseology, 6) Grammar, and 7) Conventions. Understanding these patterns may help assessment programs identify essays that need additional review and clarify whether scoring consistency is more difficult to achieve.

The supervised analysis predicts whether disagreement occurs using 1) Logistic Regression, 2) Random Forest, and 3) Gradient Boosting with 1) structured features and 2) TF-IDF text representations. Model performance is evaluated through cross-validation, hyperparameter tuning, prompt-based GroupKFold, unordered-rater-pair GroupKFold, and a locked holdout/test data set. The unsupervised analysis retains the original rating differences and uses PCA and K-means clustering to identify recurring multidimensional disagreement profiles and examine whether they can be replicated in the holdout/test sample.

A distinctive contribution of this study is its integration of disagreement prediction, multidimensional pattern discovery, and generalization testing across unfamiliar prompts and rater-pair configurations. Results show that disagreement is weakly predictable, with rater-context variables providing the strongest information. Performance remained relatively stable across unfamiliar prompts but declined toward chance level for unseen rater-pair configurations. The unsupervised analysis identified four recurring but overlapping disagreement profiles. These findings suggest that human-rater disagreement is structured but difficult to predict, particularly beyond familiar rater configurations.

Related Work

Prior research has examined human-rater disagreement in relation to essay characteristics, scoring difficulty, linguistic features, and annotator information. Lim (2019) investigated textual characteristics associated with discrepant versus consistent ratings of L2 essays, including spelling, syntactic diversity, noun phrase density, negation, voice strength, conceptual cohesion, and apparent essay length. Using discriminant function analysis, the study identified seven predictive textual features and achieved 94.4% classification accuracy on the analyzed sample. The current study extends this approach by modeling disagreement separately across seven scoring dimensions and evaluating performance on an independent holdout set and through grouped cross-validation by prompt and unordered rater pair.

Wolfe, Song, and Jiao (2016) examined whether essay features, including length, lexical diversity, sentence complexity, and spelling accuracy, were associated with scoring difficulty. They found that essay length and lexical diversity explained approximately 25% of the variance in ease of scoring. Unlike their focus on scoring accuracy relative to expert consensus, the current study directly models disagreement between paired human ratings and examines its patterns through supervised prediction and unsupervised analysis.

Wan, Kim, and Kang (2023) investigated whether disagreement among human annotators could be predicted from the text being evaluated and the annotators' demographic characteristics across five NLP datasets, including Wikipedia Politeness. They found that incorporating annotator demographics improved disagreement prediction in certain settings. Building on this approach, the current study examines paired human-rater disagreement in educational writing assessment using rater identities and evaluates predictive generalization to unseen rater-pair configurations.

Zhao and Wu (2022) examined disagreement in raters' evaluations of authorial voice, which refers to how writers express their individual perspectives, attitudes, and presence in their writing. They found that only two of 62 linguistic indices, such as measuring the frequency and variety of expressions of writer attitude, significantly distinguished essays with consistent versus discrepant voice ratings. The current study extends this work by comparing linguistic, assessment-context, student-background, rater-context, and TF-IDF features and examining disagreement patterns across one holistic and six analytic scoring dimensions.

Data Source(s)

The analysis uses the English Language Learner Insight, Proficiency, and Skills Evaluation (ELLIPSE) writing-assessment dataset, which Crossley et al. (2023) made publicly available through its GitHub repository (https://github.com/scrosseye/ELLIPSE-Corpus). The source data were provided as CSV files. Crossley et al. examined inter-rater reliability and applied Many-Facet Rasch Measurement (MFRM) to evaluate rater severity, scale characteristics, and essay fit. After excluding essays that did not meet their fit criteria, their final analysis corpus contained 6,482 essays. For the current study, merging the student data and rater files produced a working dataset of 5,840 essays and 75 columns. The original "set" variable was retained and defines the locked split of 3,510 training cases and 2,330 holdout/test cases. Each essay contains two human ratings for the seven scoring dimensions described above. Other key variables include prompt, grade, gender, race/ethnicity, socioeconomic status (SES), rater identifiers (Rater_1 and Rater_2), essay text, and linguistic measures such as number of words, sentences, paragraphs, MTLD, and TTR. The dataset includes 27 rater codes, and all 44 prompts are represented in both the training and holdout/test sets.

Table 1
Data Source(s)

Dataset property	Value
Analysis sample	5,840 essays
Training / Test (Holdout)	3,510 / 2,330 (locked by original `set` variable)
Scoring dimensions	Overall, Cohesion, Syntax, Vocabulary, Phraseology, Grammar, Conventions
Raters	27 rater codes in both training and holdout
Prompts	44 prompts represented in both training and holdout
Text field	QA_Full_Text_main

Feature Engineering

The first step was to preserve the paired ratings and derive disagreement outcomes. For each scoring dimension d (e.g., Grammar), the absolute paired-rating difference was first calculated as:

$$\Delta_d = |Score_{2d} - Score_{1d}|$$

The primary supervised target then distinguished **exact agreement from disagreement**:

$$Y_d = \begin{cases} 0, & \text{if } \Delta_d = 0, \\ 1, & \text{if } \Delta_d > 0. \end{cases}$$

Thus, while (Y_d = 1) indicated disagreement between the two raters, (Y_d = 0) indicated exact agreement.

For the unsupervised analysis, the seven original Δ_d variables defined above were analyzed together to capture which scoring dimensions showed disagreement between the two raters and the magnitude of those disagreements for each essay. The same seven variables were used for both the full and disagreement-only samples; the only difference is that the disagreement-only sample excludes essays with (Delta_d = 0) across all seven dimensions.

Part A. Supervised Learning

Methods

Study Question Formulation and Data Split. The supervised analysis modeled paired human-rater disagreement separately across seven scoring dimensions, using the absolute difference between two human ratings for each dimension. The pre-existing dataset split was preserved, with 3,510 essays for model development and 2,330 as an untouched holdout/test set. All model comparisons, feature selection, hyperparameter tuning, and robustness analyses were conducted using training data before final holdout evaluation. Human scores used to define disagreement were excluded from predictors to prevent target leakage. This design examined whether disagreement could be predicted from assessment context, essay features, student background, rater information, and essay text.

Feature Representations and Preprocessing. Multiple feature representations were compared to determine which sources of information were most effective for predicting disagreement. Structured representations were evaluated with 1) Logistic Regression, 2) Random Forest, and 3) Gradient Boosting, while TF-IDF representations were evaluated with 1) Logistic Regression only.

Table 2
Feature Representations for Evaluation

Representation	Predictors / construction	Rationale
Assessment context	prompt, grade	Prompt and graded context
Linguistic / writing	num_words, num_sent, num_para, MTLD, TTR	Essay length, organization, and lexical characteristics
Student background	gender, race ethnicity, SES	Student's background characteristics
Rater context	Rater 1, Rater 2	Rater identities and configuration
Word TF-IDF	Essay text represented by word unigrams and bigrams (maximum 5,000 features)	Lexical content of essay text
Combined representations	context + linguistic; all structured; all structured + TF-IDF	Testing whether combining information sources improves prediction

Missing numeric values were imputed using the median value of each variable, and numeric predictors were standardized. Missing categorical values were imputed using the most frequent category

of each variable, followed by one-hot encoding with handling for unseen categories. TF-IDF used min_df = 3, max_df = 0.95, word unigrams and bigrams, a maximum of 5,000 features, and sublinear term-frequency scaling. All preprocessing steps, including TF-IDF vocabulary construction, were based only on the training data within each cross-validation fold to prevent information leakage.

Model Families and Hyperparameter Tuning. Three model families were used to compare linear and nonlinear approaches to disagreement prediction. While Logistic Regression served as a regularized linear baseline, Random Forest and Gradient Boosting captured nonlinear relationships and interactions through different tree-based ensemble strategies. Structured features were evaluated with all three models, but TF-IDF text-only and TF-IDF-plus-structured representations were evaluated with Logistic Regression only. Random Forest combines independently fitted trees, while Gradient Boosting builds trees sequentially to correct earlier errors.

Table 3

Model Families and Hyperparameter Exploration

Model	Role	Hyperparameter grid
Logistic Regression	Regularized linear baseline; tests whether disagreement can be captured by linear predictor effects.	C = 0.01, 0.1, 1, 10
Random Forest	Bagged randomized trees; captures nonlinearities and interactions while reducing variance across trees.	n_estimators = 300, 500; max_depth = None, 10, 20; min_samples_leaf = 1, 2, 5; max_features = sqrt, 0.5
Gradient Boosting	Sequential tree ensemble; captures nonlinear patterns by iteratively correcting errors from earlier trees.	n_estimators = 100, 200; learning_rate = 0.03, 0.05, 0.1; max_depth = 2, 3; subsample = 0.8, 1.0

Note. TF-IDF text-only and TF-IDF-plus-structured representations were evaluated with Logistic Regression only.

Model and feature combinations were first compared using 5-fold stratified cross-validation, with mean ROC-AUC as the primary selection metric. The best configuration for each of the seven outcomes underwent hyperparameter tuning using training data only. Robustness was evaluated using GroupKFold by unordered rater pair and by prompt. After all model choices were finalized, the seven selected models were fitted on the full training set and evaluated on the untouched holdout set.

Supervised Learning Evaluation

Evaluation Metrics. ROC-AUC was used as the primary evaluation metric because it measures how well the models distinguish agreement from disagreement across classification thresholds and provides a consistent basis for comparison across the seven outcomes. Model comparisons were summarized using mean 5-fold cross-validation ROC-AUC ± SD. Average Precision (AP)/PR-AUC, balanced accuracy, F1, sensitivity, and specificity were also examined as complementary performance measures.

Overall Results. Table 4 summarizes the best-performing feature representation within each model family across the seven scoring outcomes. The highest-performing model–feature combination across all three families was then selected for each outcome. Predictive discrimination was weak, with mean 5-fold cross-validation ROC-AUC ranging from .525 to .553 and only small differences among model families. Rater-only features performed best in most comparisons, while all-structured features performed best less often, and neither TF-IDF representation was best. For example, for Cohesion, all

three model families performed similarly (ROC-AUC = .552–.553), with rater-only features performing best in each.

Table 4
Best Result within Each Model Family (5-Fold CV ROC-AUC Mean +/- SD)

Outcome	Logistic Regression	Random Forest	Gradient Boosting
Cohesion	0.552 +/- 0.020 (R)	0.553 +/- 0.017 (R)	0.552 +/- 0.021 (R)
Conventions	0.527 +/- 0.011 (R)	0.533 +/- 0.016 (R)	0.531 +/- 0.015 (R)
Grammar	0.540 +/- 0.022 (AS)	0.536 +/- 0.011 (R)	0.548 +/- 0.013 (R)
Overall	0.533 +/- 0.020 (AS)	0.536 +/- 0.010 (R)	0.538 +/- 0.011 (R)
Phraseology	0.537 +/- 0.015 (R)	0.529 +/- 0.012 (R)	0.537 +/- 0.019 (AS)
Syntax	0.525 +/- 0.019 (AS)	0.533 +/- 0.008 (AS)	0.534 +/- 0.022 (R)
Vocabulary	0.529 +/- 0.023 (R)	0.536 +/- 0.029 (R)	0.527 +/- 0.022 (R)

Note. R = rater_only; AS = all_structured. The table reports the strongest representation within each algorithm family for each outcome.

Selected Models, Tuning, Robustness, and Holdout Performance. The rater-only representation was selected for all seven outcomes. Random Forest was selected for Cohesion, Conventions, and Vocabulary; Gradient Boosting for Grammar, Overall, and Syntax; and Logistic Regression for Phraseology. Hyperparameter tuning produced only small improvements over the initial cross-validation results. Performance remained relatively stable when entire prompts were held out but declined toward chance when entire unordered rater pairs were excluded from training. Holdout ROC-AUC ranged from approximately .513 to .566, with Cohesion highest and Overall lowest, broadly consistent with conventional cross-validation. Across outcomes, prompt GroupKFold differed from tuned CV by only about −.003 ROC-AUC on average, whereas rater-pair GroupKFold was lower by approximately .036. This decline occurred across all seven dimensions and fell below .50 for Conventions, indicating that predictive performance was much more sensitive to unfamiliar rater-pair configurations than to unfamiliar prompts.

Table 5
Selected Configurations and Validation-Regime Performance

Outcome	Model	Initial CV	Tuned CV	Prompt GKF	Rater-pair GKF	Holdout ROC-AUC
Cohesion	RF	0.553 +/- 0.017	0.5570	0.5556 +/- 0.0117	0.5077 +/- 0.0280	0.5663
Conventions	RF	0.533 +/- 0.016	0.5345	0.5325 +/- 0.0207	0.4872 +/- 0.0177	0.5609
Grammar	GB	0.548 +/- 0.013	0.5510	0.5446 +/- 0.0139	0.5142 +/- 0.0310	0.5457
Overall	GB	0.538 +/- 0.011	0.5377	0.5266 +/- 0.0203	0.5045 +/- 0.0235	0.5128
Phraseology	LR	0.537 +/- 0.015	0.5401	0.5430 +/- 0.0268	0.5101 +/- 0.0306	0.5497
Syntax	GB	0.534 +/- 0.022	0.5386	0.5392 +/- 0.0191	0.5091 +/- 0.0397	0.5260
Vocabulary	RF	0.536 +/- 0.029	0.5369	0.5342 +/- 0.0114	0.5106 +/- 0.0290	0.5462

Note. LR = Logistic Regression; RF = Random Forest; GB = Gradient Boosting; CV = cross-validation; GKF = GroupKFold. Initial CV and grouped-validation results are reported as mean ROC-AUC ± SD; tuned CV as mean ROC-AUC; and holdout performance as a single ROC-AUC value.

In-Depth Evaluation

Deeper diagnostic analyses focused on the Cohesion disagreement model, which achieved the highest tuned cross-validation ROC-AUC (.5570) among the seven outcomes (see Table 5). The selected model used rater-only features and Random Forest with max_depth = 10, max_features = 0.5, min_samples_leaf = 5, and n_estimators = 300. Model selection was based exclusively on training-development results.

Feature Importance and Ablation Analysis. Feature importance was examined using permutation analysis, which measured how much validation ROC-AUC decreased when each predictor was disrupted during cross-validation. As shown in Table 6, both rater-position variables contributed to the Cohesion model. Rater_1 produced a larger average decrease in ROC-AUC, but its contribution also varied more across cross-validation folds. Thus, Rater_1 appeared more influential on average, although the results do not show that it was consistently more important than Rater_2.

Table 6

Cross-Validated Permutation Feature Importance for the Cohesion Model

Feature	Mean ROC-AUC decrease	SD ROC-AUC decrease
Rater_1	0.041304	0.025099
Rater_2	0.031208	0.004731

Raw-feature ablation reinforced the permutation-importance results by testing what happened when each rater predictor was removed and the model was refit without it (see Table 7). The full rater-only feature set achieved a mean CV ROC-AUC of .5570 and served as the baseline for comparison. While removing Rater_1 reduced ROC-AUC to .5395, corresponding to an ROC-AUC loss of .0175, removing Rater_2 reduced it to .5424, an ROC-AUC loss of .0146. Thus, the model performed better when both rater variables were included, with removing Rater_1 producing a slightly larger decrease in ROC-AUC than removing Rater_2.

Table 7

Best-Model Raw-Feature Ablation

Condition	Features used	ROC-AUC mean	ROC-AUC SD	ROC-AUC loss
Without Rater 1	Rater 2	0.5395	0.0155	0.0175
Without Rater 2	Rater 1	0.5424	0.0276	0.0146
Full rater only feature set	Rater 1, Rater 2	0.5570	0.0199	0.0000

Supplementary Feature-Family Ablation Analysis. To examine whether broader groups of predictors added information beyond rater context, an additional analysis used the same tuned Cohesion Random Forest but started from the full structured feature representation. One feature family was removed at a time, and the model was reevaluated using 5-fold cross-validation (see Table 8). The full structured model achieved a mean CV ROC-AUC of .5187 and served as the baseline. Removing rater context produced the largest performance loss, reducing ROC-AUC to .4931, and removing assessment context produced only a small loss. In contrast, removing student-background or linguistic/writing features slightly improved performance. These results provide additional evidence that rater context supplied the strongest predictive information among the structured feature families.

Table 8

Supplemental Feature-Family Ablation for Cohesion

Condition	ROC-AUC mean	ROC-AUC SD	ROC-AUC loss when removed
Without rater context	0.4931	0.0296	0.0256
Without assessment context	0.5151	0.0179	0.0036
Full all structured	0.5187	0.0134	0.0000
Without student background	0.5221	0.0187	-0.0034
Without linguistic/writing	0.5351	0.0157	-0.0163

Sensitivity Analysis. Two sensitivity analyses were conducted. First, the Cohesion Random Forest model was evaluated using all 36 hyperparameter configurations specified in Table 3. Across these 36 configurations, mean cross-validation ROC-AUC ranged from .5525 to .5570, a difference of only .0045. Table 9 shows the five best-performing configurations from this set of 36, with mean ROC-AUC ranging from .5565 to .5570. All five used min_samples_leaf = 5. Overall, these results indicate that the model's predictive performance changed very little across the 36 tested hyperparameter configurations.

Table 9

Top Five Cohesion Random Forest Hyperparameter Configurations

Rank	max_depth	max_features	min_leaf	n_estimators	Mean CV ROC-AUC	SD
1	10	0.5	5	300	0.5570	0.0178
2	20	0.5	5	300	0.5569	0.0155
3	20	sqrt	5	500	0.5568	0.0173
4	10	sqrt	5	300	0.5567	0.0168
5	None	sqrt	5	500	0.5565	0.0160

Second, sensitivity to the validation structure was examined across all seven scoring outcomes (see Table 5). Prompt-grouped cross-validation produced ROC-AUC values close to conventional tuned cross-validation, but rater-pair-grouped ROC-AUC declined across all seven outcomes. These findings suggest that the models generalized less successfully to unseen rater-pair configurations than to unseen writing prompts.

Tradeoffs. The Cohesion model illustrates the tradeoff between sensitivity and specificity at different classification thresholds (see Table 10). While the default threshold was .50, alternative thresholds were examined descriptively to assess their effects on disagreement detection and false-positive classifications. Lower thresholds increased sensitivity but also increased false positives, whereas higher thresholds improved specificity but missed more actual disagreements. Because the holdout set had already been used for final evaluation, this analysis was descriptive only, and no alternative threshold was selected or independently validated.

Table 10

Cohesion Threshold Tradeoffs on the Holdout Data Set

Threshold	Precision	Sensitivity	Specificity	F1	Balanced accuracy	Flagged rate
0.45	0.490	0.858	0.199	0.624	0.528	0.828
0.50	0.535	0.413	0.678	0.466	0.545	0.365
0.55	0.551	0.230	0.831	0.325	0.531	0.198
0.60	0.556	0.131	0.906	0.212	0.519	0.111

At the default .50 threshold, sensitivity was .413 and specificity was .678, with 36.5% of holdout essays flagged as predicted disagreements. Lowering the threshold to .45 increased sensitivity to .858 but also increased the flagged rate to 82.8%. On the other hand, raising the threshold to .55 increased specificity to .831, but it reduced sensitivity to .230. At the highest threshold of .60, specificity increased further to .906, but sensitivity declined to .131. None of the four thresholds achieved both high sensitivity and high specificity. These findings suggest that adjusting the classification threshold alone cannot overcome the model's limited ability to distinguish between agreement and disagreement cases.

Failure Analysis. Three incorrectly predicted holdout essays were selected to illustrate different types of model errors (see Table 11). The total number of training essays jointly scored by each corresponding rater pair was also included in Table 11 to provide context for interpreting these errors.

Table 11

Three Specific Cohesion Holdout Failure Examples

Failure category	Row	Actual / Prediction	P (disagree)	Rater pair (train n)	Prompt / grade	Cohesion scores
False negative	4380	1 / 0	0.330	rater_4 || rater_6 (19)	Distance learning / 12	3 vs 4
False positive	3790	0 / 1	0.654	rater_23 || rater_9 (11)	Impact of technology / 11	3 vs 3
Low-frequency rater-pair false negative	4310	1 / 0	0.457	rater_16 || rater_20 (3)	Creative arts requirement / 12	3 vs 4

Note. Rater-pair training counts indicate the number of essays jointly scored by each pair in the training sample.

In the first example, the two raters assigned different Cohesion scores (3 and 4), but the model incorrectly predicted agreement, even though the same rater pair had scored 19 essays during training. The second example illustrates a false positive, in which the model predicted disagreement despite both raters assigning a score of 3. In the third example, the model incorrectly predicted agreement for a rater pair represented by only three training essays. These examples illustrate that the model can incorrectly predict disagreement even for rater pairs previously observed during training. The error involving a less frequently observed rater pair is also consistent with the broader finding that prediction performance declined for unseen rater-pair configurations.

Future improvements could test richer essay-level features alongside rater-context information and increase representation of infrequently observed rater pairs. These changes could help determine whether prediction can rely less on familiar rater configurations and generalize better to sparse or unseen rater pairs.

Supervised Learning Summary

Across seven holistic and analytic scoring dimensions, supervised models showed weak but reproducible discrimination under conventional cross-validation and final holdout evaluation. Rater-context features consistently provided the strongest signal, tuning gains were small, and prompt-grouped performance remained comparatively stable. In contrast, performance declined toward chance level when entire unordered rater-pair configurations were held out. The deeper evaluation with Cohesion supported the same conclusion from several directions: permutation importance and ablation concentrated predictive value in the two rater variables, performance was stable across reasonable hyperparameter settings, threshold choice involved a substantial sensitivity-specificity tradeoff, and failure cases illustrated the limitations of relying primarily on rater-context information. Overall, the supervised results suggest that disagreement is structured but only weakly predictable, with predictive performance strongly dependent on rater configuration.

Part B. Unsupervised Learning

The unsupervised analysis examined whether paired human-rater disagreement across the seven holistic and analytic scoring dimensions could be represented by a smaller set of dimensions and whether recurring multidimensional disagreement profiles could be identified. Results are organized by methods, solution and evaluation, holdout replication, and additional exploratory analysis.

Methods

The original Delta variables used for the supervised learning were retained because they share the same score-difference scale and preserve the magnitude of disagreement, and the unsupervised analysis used both the full training sample and the disagreement-only sample. The analyses utilized both Principal Component Analysis (PCA) and K-means clustering on the seven absolute paired-rating differences. While PCA was used to examine whether disagreement across the seven scoring dimensions could be represented by fewer underlying components, K-means was used to identify recurring multidimensional disagreement profiles. K-means solutions from ($k = 2$) through ($k = 8$) were compared using silhouette score, cluster-size balance, random-initialization stability, and centroid interpretability. All solution selection was completed using training data before applying the frozen solution to the holdout/test sample.

Analysis Samples. Two samples were used for the unsupervised analysis. The full sample included all 5,840 essays, comprising 3,510 training and 2,330 holdout essays, regardless of whether the raters agreed or disagreed. The disagreement-only sample included essays with at least one difference between the two raters' scores across the seven scoring dimensions. This sample contained 5,700 essays (3,425 training and 2,275 holdout), excluding 140 essays (85 training and 55 holdout) for which the two raters agreed on all seven dimensions.

Evaluation Metrics. PCA was evaluated using explained variance and component loadings to assess dimensional reduction and interpret the resulting components. K-means was evaluated using silhouette score, cluster-size balance, ARI across random initializations, and centroid patterns to assess cluster separation, stability, and substantive interpretability.

Unsupervised Solution and Evaluation

PCA Results. PCA showed that disagreement was multidimensional rather than dominated by a single factor. In the full training sample, PC1 explained 19.2% of the variance, the first three components explained 49.0%, and the first six components explained about 89.9%. Similar results were observed in the disagreement-only training sample: PC1 explained 17.8%, the first three components explained 48.2%, and the first six components explained 89.9% (see Figure 1).

The loading patterns supported this conclusion (see Figure 2). In the full sample, Overall and Cohesion contributed most strongly to PC1, while later components were more strongly associated with Grammar, Conventions, Cohesion, Syntax, and Vocabulary. The disagreement-only PCA showed the same overall pattern of variance distributed across several components.

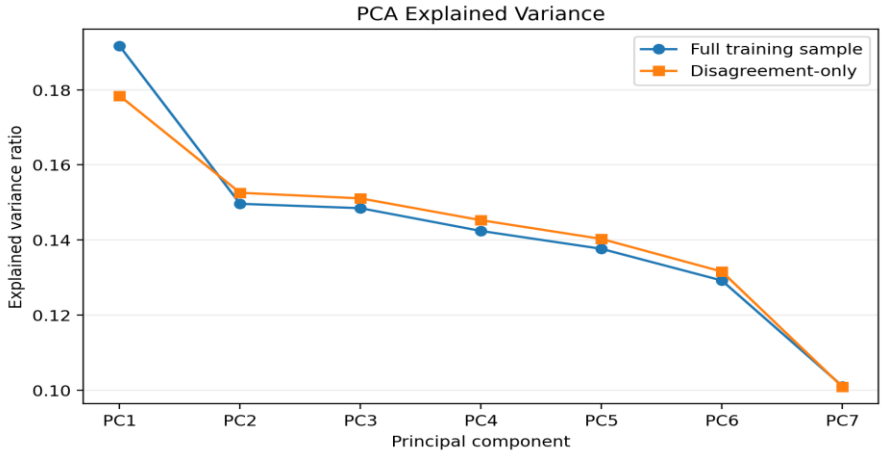


Figure 1. Explained variance ratios for PCA in the full and disagreement-only training samples.

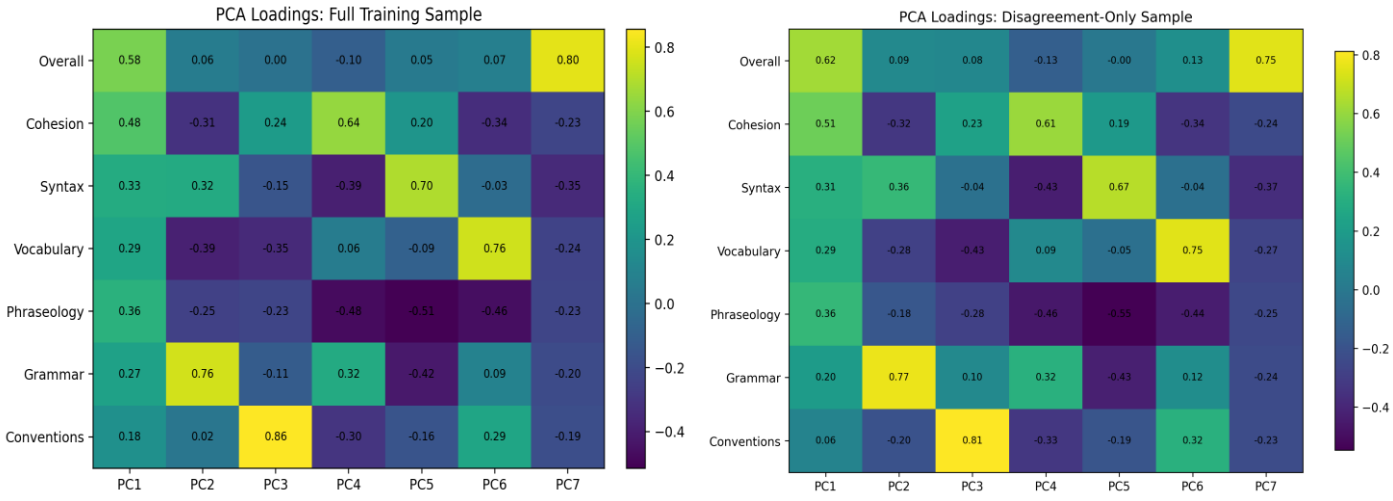


Figure 2. *PCA loadings for the full training sample and disagreement-only sample.*

K-means Solution Comparison. Solutions from (k = 2) through (k = 8) were compared using silhouette score, cluster size, stability, and interpretability (see Table 12 and Figure 3). In both the full and disagreement-only training samples, (k = 4) produced the highest silhouette score (.168 and .169,

respectively) and also yielded relatively even cluster sizes. Considering these results together with stability and interpretability, the four-cluster solution was retained for both analyses.

Table 12

K-Means Solution Comparison across Candidate Values of k

k	Full silhouette	Full cluster n range	Disagree-only silhouette	Disagree-only cluster n range
2	0.153	1750-1760	0.151	1675-1750
3	0.144	941-1356	0.143	948-1375
4	0.168	801-969	0.169	801-924
5	0.145	497-846	0.156	529-824
6	0.149	500-692	0.161	444-809
7	0.163	404-676	0.148	410-580
8	0.161	267-512	0.160	342-508

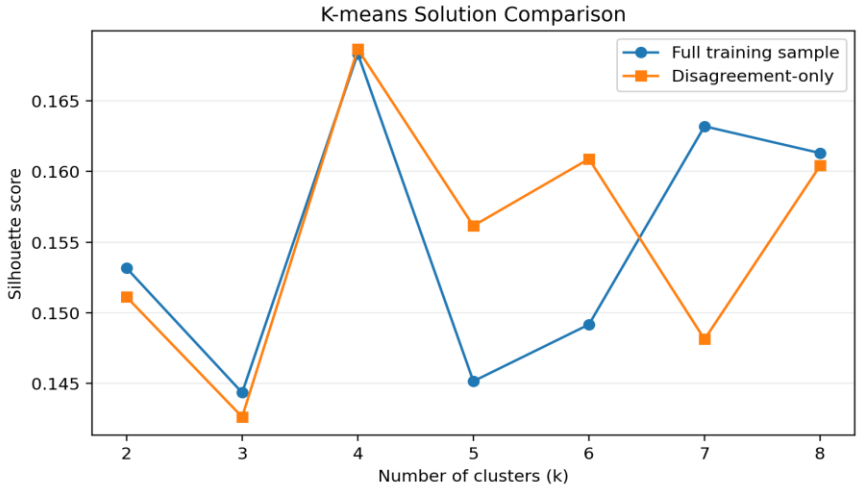


Figure 3. Silhouette scores for candidate K-means solutions.

Sensitivity and Stability. Random-initialization sensitivity was examined across two to eight clusters (see Figure 4). The two- and three-cluster solutions generally showed higher mean pairwise Adjusted Rand Index (ARI) values, indicating greater consistency across random starts than the four-cluster solution. At $k = 4$, mean pairwise ARI was .400 (minimum = .170, maximum = .840, SD = .209) for the full training sample and .231 (minimum = .088, maximum = .629, SD = .149) for the disagreement-only sample (see Table 13). Despite this limited stability, the four-cluster solution was retained based on its balance of cluster separation, size, and substantive interpretability (see Table 12). The low silhouette scores (.168 and .169; Table 12) indicate that the four clusters were not clearly separated, while the modest ARI values (.400 and .231; Table 13) indicate that repeating K-means with different random starting points did not consistently produce the same groupings of essays.

Table 13

Stability of the Four-Cluster K-means Solution across Random Initializations

k	Mean pairwise ARI	Minimum pairwise ARI	Maximum pairwise ARI	SD pairwise ARI	Sample
4	0.400	0.170	0.840	0.209	Full Training Sample
4	0.231	0.088	0.629	0.149	Disagreement-only Sample

Note. ARI = Adjusted Rand Index.

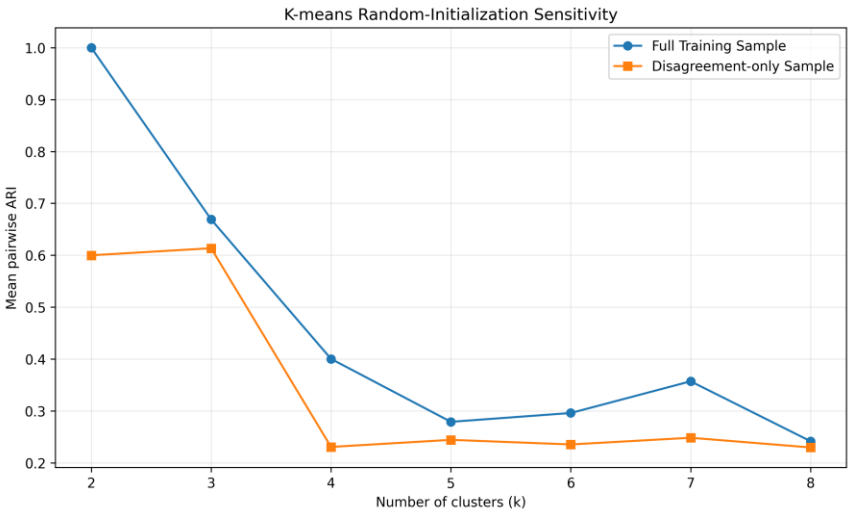


Figure 4. K-means cluster stability across random initializations (k = 2–8).

Four-Profile K-means Solution. Figure 5 presents heatmaps for both the full and disagreement-only training samples, but substantive interpretation focused on the 3,425 disagreement-only essays. The *k* = 4 solution identified four interpretable but overlapping disagreement profiles (Figure 5 and Table 14). Profile 1 (25.8%), distributed language-trait disagreement, showed moderate disagreement across Syntax, Vocabulary, Phraseology, and Grammar, with little Cohesion or Conventions disagreement. Profile 2 (23.8%), broad multidimensional disagreement, showed the widest pattern, including mean disagreement of approximately one point in Overall and Cohesion. Profile 3 (27.0%), cohesion-centered disagreement, showed mean Cohesion disagreement of approximately one point with little Overall disagreement. Profile 4 (23.4%), conventions-centered disagreement, showed mean Conventions disagreement of approximately one point with little Cohesion disagreement. It should be emphasized that although K-means was fitted separately to the two training samples, both solutions showed similar disagreement patterns, with profile numbers differing between the heatmaps.

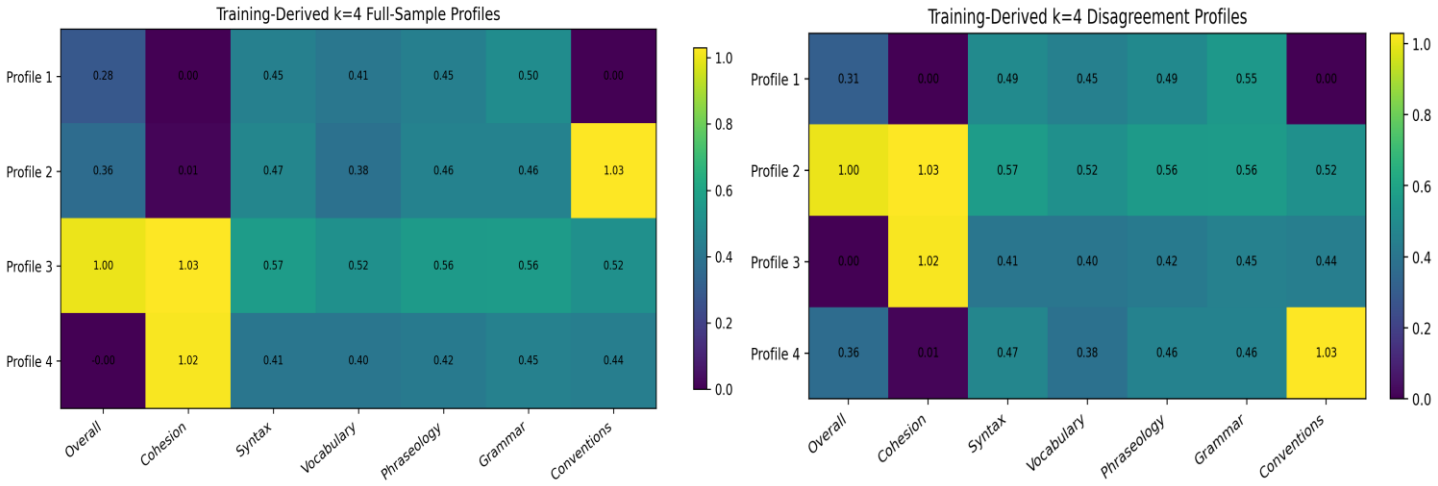


Figure 5. Training-derived centroids for the four full-sample and disagreement-only K-means profiles.

Table 14

Interpretation of the Four Disagreement-only Profiles

Profile	Interpretive label	Training n (%)	Defining pattern
1	Distributed language-trait disagreement	884 (25.8%)	No Cohesion/Conventions disagreement; moderate Syntax, Vocabulary, Phraseology, Grammar
2	Broad multidimensional disagreement	816 (23.8%)	Broad disagreement; Overall and Cohesion ≈ 1 point
3	Cohesion-centered disagreement	924 (27.0%)	Cohesion ≈ 1 point; Overall ≈ 0
4	Conventions-centered disagreement	801 (23.4%)	Conventions ≈ 1 point; Cohesion ≈ 0

Holdout Replication

The four disagreement profiles established during training were applied to the 2,275 disagreement-only holdout essays. Each holdout essay was assigned to its closest training-derived cluster without refitting K-means. All four profiles were represented in the holdout sample, with proportions closely resembling those in training (see Table 15). The largest difference was 2.1 percentage points for Profile 1 (25.8% in training versus 27.9% in holdout). The training and holdout centroids also showed small observed Euclidean distances, ranging from .044 to .096. Profile 3 had the smallest distance (.044), followed by Profile 1 (.070), Profile 4 (.082), and Profile 2 (.096). Together, the similar profile proportions and small centroid differences provide descriptive evidence that the training-derived disagreement patterns were also observed in the holdout sample, despite modest internal cluster separation.

Table 15

Holdout Replication of the Disagreement-Only Four-Profile Solution

Profile	Training (%)	Holdout (%)	Difference (%)	Centroid distance
1	25.8	27.9	+2.1	0.070
2	23.8	22.9	-0.9	0.096
3	27.0	25.3	-1.7	0.044
4	23.4	23.9	+0.5	0.082

Additional Exploratory Analysis

The supervised analysis showed that rater-context variables provided the strongest predictive information, and the unsupervised analysis identified four recurring disagreement profiles across the seven scoring dimensions. Based on these findings, an exploratory analysis examined whether disagreement-profile membership was associated with rater-pair configurations in the training sample and whether a similar association appeared in the holdout sample (see Table 16). A chi-square test was used to assess the association, Cramer's V to measure its strength, and a permutation test to provide an additional significance assessment because many expected cell counts were below five. Standardized residuals were examined descriptively to identify specific rater-pair and profile combinations occurring more or less frequently than expected.

In the disagreement-only training sample, 3,425 essays were scored by 72 distinct rater pairs, defined as unique combinations of two raters regardless of scoring position. The association between rater-pair configuration and profile membership was statistically significant, $\chi^2(213) = 279.82$, $p = .0014$, with Cramer's V = .165; a 2,000-permutation test was also significant ($p = .001$). Among rater pairs represented by at least 20 essays, 13 pair-profile cells had absolute standardized residuals of at least 2.0, indicating localized over- or under-representation. For example, rater_12 || rater_19 was over-represented in Profile 4 (residual = 2.93), rater_12 || rater_15 was under-represented in Profile 1 (−2.56), and rater_1 || rater_2 was over-represented in Profile 2 (2.39). Among sufficiently represented rater pairs, these localized departures suggest that the overall association was concentrated in a limited number of rater-pair and profile combinations rather than uniformly across all pairs. A similar overall association was observed among the 2,275 disagreement-only holdout essays, $\chi^2(213) = 270.04$, $p = .0049$, with Cramer's V = .199; the permutation test was also significant ($p = .002$). These findings provide exploratory evidence that disagreement-profile membership varied across rater-pair configurations in both samples, although the associations were modest in magnitude.

Table 16

Rater-Pair by Disagreement-Profile Association in Training and Holdout Samples

Sample	n essays	Pairs	χ^2(df)	χ^2 p	Cramer's V	Permutation p
Training	3,425	72	279.82 (213)	.0014	.165	.001
Holdout	2,275	72	270.04 (213)	.0049	.199	.002

Unsupervised Learning Summary

Overall, PCA and K-means provided complementary insights into the structure of paired human-rater disagreement. While PCA showed that no single component dominated the variance, K-means identified four recurring but overlapping disagreement profiles across the seven scoring dimensions. Although cluster separation and random-start stability were modest, the profiles showed similar proportions and centroid patterns in the holdout sample. Additional exploratory analysis also found modest associations between profile membership and rater-pair configurations in both samples. These findings suggest that human-rater disagreement is multidimensional, exhibits reproducible patterns, and is partly associated with rater-pair configuration.

Conclusion and Discussion

The supervised analysis showed that paired human-rater disagreement was weakly predictable from the available features, with cross-validation and holdout ROC-AUC values ranging from .513 to .566. While performance remained relatively stable across unseen prompts, it declined toward chance level for unseen rater-pair configurations. These findings suggest that predictive information depended more on familiar rater configurations than on transferable essay characteristics. Further evaluation of the best-performing Cohesion model also indicated that while rater-context variables contributed the strongest predictive information, student-background and linguistic features added little. Performance varied minimally across Random Forest hyperparameter settings, indicating that weak prediction was unlikely due primarily to poor tuning. Finally, threshold analysis highlighted the tradeoff between detecting actual disagreements and incorrectly flagging agreements, and failure analysis showed that the

model could make errors even for rater pairs observed during training. The unsupervised analysis provided a complementary view of human-rater disagreement. PCA showed that no single component dominated the variation across the seven scoring dimensions, indicating that disagreement was multidimensional rather than a single general tendency. K-means analysis identified four recurring but overlapping disagreement profiles among disagreement-only essays. Although cluster separation was modest, the four profiles were reproduced closely in the holdout sample in both their prevalence and overall centroid patterns. Thus, while individual disagreement was difficult to predict, disagreement that occurred showed recurring multidimensional forms.

The most surprising supervised finding was that while performance remained relatively stable across unfamiliar prompts, it substantially declined for unfamiliar rater-pair configurations. This difference was also identified through grouped cross-validation, highlighting limited generalization to unseen rater-pair configurations. One surprising unsupervised finding was that the four disagreement profiles identified in the training sample showed similar patterns in the holdout/test sample, despite modest cluster separation and random-start stability. This presented a challenge in interpreting the clusters without overstating their distinctness. Thus, the profiles were considered as overlapping patterns rather than natural classes.

Future studies could examine how particular rater combinations are associated with recurring disagreement patterns and whether this information could support additional review or rater training. Richer essay representations, such as semantic content, organization and discourse, may help identify scoring difficulties beyond rater identity. Future studies could also evaluate disagreement screening by selecting thresholds on development data and validating them on an independent sample. Finally, future studies could test whether similar disagreement profiles appear in other writing-assessment datasets and whether they remain similar when different clustering methods are used.

Ethical Considerations

The supervised models predict whether two human raters will disagree on an essay, but their predictive performance is weak. They should be used only as exploratory decision-support tools, not for high-stakes decisions. Although rater-context variables provide the strongest predictive information, this does not mean that raters or rater pairs are inherently problematic. Rater identities should remain anonymous, and the findings should not be used for personnel decisions without independent review. Background variables, such as gender, race/ethnicity, and socioeconomic status, also require careful use. Because the dataset contains student writing, scores, background information, and rater identifiers, privacy protections are essential. The unsupervised analysis identifies disagreement profiles across the seven scoring dimensions, but these remain exploratory patterns. Associations between profiles and rater pairs or other characteristics should not be treated as causal or as evidence that any group is problematic. Because the clusters showed modest separation and some sensitivity to initialization, assigning individual essays to disagreement profiles should be cautiously interpreted. The profiles are best used to support research and further investigation. Future use should include human review and validation in new samples.

Statement of Work

I, Daeryong Seo, completed all aspects of the project.

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